// Chenyu Pan

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Portfolio

Chenyu Pan

EDUCATION

University of Waterloo – BASc, Electrical Engineering | **Sep 2025 – Apr 2030**

SKILLS

Hardware: Schematic Capture, PCB Layout, Power Electronics, Embedded Hardware, Simulation and Prototyping
Software and Protocols: Altium, KiCad, LTspice, C, C++, Python, Verilog, Git; I2C, SPI, UART, CAN, LIN, USB
Lab Equipment: Oscilloscope, Signal Generator, Power Supply, E-Load, Logic Analyzer, DMM, Soldering and Rework

EXPERIENCE

Systems Engineering Intern – WSP Canada – Toronto, ON | **Jan 2026 – Apr 2026**

- Constructed a vector-based modeling engine in C# for Traction Power, simulating physical strain and feasibility of overhead contact power lines for proposed Toronto LRT projects, reducing the associated development duration by 70%
- Applied RF propagation modelling to calibrate train control antenna systems, managing path loss to exceed the noise floor
- Designed a Python and C# tool for the Union Station Enhancement Project to generate requirement matrices and data dashboards from IBM DOORS modules, saving time by 80% and allowing junior engineers to snapshot baselines
- Developed an automated Python script to parse large system datasets into CSVs, cutting turnaround time by 90%
- Maintained a comprehensive security traceability matrix and formulated APTA risk reports for the Ontario Line's metro stations, ensuring that all technical changes and requirements complied with risk mitigations and safety standards

Electrical Team Member – Waterloo Aerial Robotics Group – Waterloo, ON | **Sep 2025 – Present**

- Developed a stable 3.3V LDO regulator in Altium, achieving <1% voltage ripple under high-di/dt transient loads
- Installed wire harnesses with the mechanical team to withstand EMI, signal integrity, and high-vibration settings
- Performed system integration, board bring-up, and signal integrity analysis using DMMs, DSOs, and Logic Analyzers for new CAN data logger and buck converter PCBAs to validate 100% data packet delivery in high-noise conditions
- Spearheaded component research for a servo driver PCBA project, reducing the Bill of Materials by 20%

Data and Technology Intern – City of Vaughan – Vaughan, ON | **Sep 2024 – Jan 2025**

- Consolidated 250+ infrastructure asset types in ArcGIS and AutoCAD and reevaluated the conditions of 200+ city roads through field work to align urban planning for Corporate Asset Management with other departments
- Integrated GIS hardware scans and Power Automate workflows, improving asset location accuracy by 95%

PROJECTS

Quasi-Resonant Flyback Power Converter

- Architected a 120VAC power conversion system in Altium with quasi-resonant topology and custom magnetics, reducing heat losses by 40%+ versus fixed flybacks with 92% efficiency at 60W while maintaining galvanic isolation
- Created an inrush precharge circuit in LTspice and validated SMPS steady state in SIMPLIS to ensure 'valley' switching

Dual-Motor Differential Drive Controller

- Designed an STM32-based BLDC differential drive controller in Altium with 3-phase motor drivers and LTspice-validated power regulation; iterated PCB revisions to improve analog signal integrity and deliver nearly 400W to motors
- Integrated communications stack with hall sensing, 6-axis IMU, MicroSD logging, and CAN / UART debugging interfaces

HawkEye – Quadcopter Flight Controller | **In Progress**

- Developing a 4S-powered STM32-based quadcopter flight controller in KiCad featuring USB-C, flash memory, redundant IMUs, a barometer, I/O peripheral expansion, and ExpressLRS RF with a dedicated ESP32 for compute-heavy UAV control